that food supplies are not over-exploited. In the condition of the experiment, it so happened that food was never going to be scarce, but the mice could not be expected to realize that. They are programmed for life in the wild, and it is likely that in natural conditions over-crowding is a reliable indicator of future famine.

What does the selfish gene theory say? Almost exactly the same thing, but with one crucial difference. You will remember that, according to Lack, animals will tend to have the optimum number of children from their own selfish point of view. If they *bear* too few or too many, they will end up *rearing* fewer than they would have if they had hit on just the right number. Now, 'just the right number' is likely to be a smaller number in a year when the population is over-crowded than in a year when the population is sparse. We have already agreed that over-crowding is likely to foreshadow famine. Obviously, if a female is presented with reliable evidence that a famine is to be expected, it is in her own selfish interests to reduce her own birth rate. Rivals who do not respond to the warning signs in this way will end up rearing fewer babies, even if they actually bear more. We therefore end up with almost exactly the same conclusion as Wynne-Edwards, but we get there by an entirely different type of evolutionary reasoning.

The selfish gene theory has no trouble even with 'epideictic displays'. You will remember that Wynne-Edwards hypothesized that animals deliberately display together in large crowds in order to make it easy for all the individuals to conduct a census, and regulate their birth rates accordingly. There is no direct evidence that any aggregations are in fact epideictic, but just suppose some such evidence were found. Would the selfish gene theory be embarrassed? Not a bit.

Starlings roost together in huge numbers. Suppose it were shown, not only that over-crowding in winter reduced fertility in the following spring, but that this was directly due to the birds' listening to each other's calls. It might be demonstrated experimentally that individuals exposed to a tape-recording of a dense and very loud starling roost laid fewer eggs than individuals exposed to a recording of a quieter, less dense, roost. By definition, this would indicate that the calls of starlings constituted an epideictic display. The selfish gene theory would explain it in much the same way as it handled the case of the mice.

Again, we start from the assumption that genes for having a larger family than you can support are automatically penalized, and become